

Section 7.2 Exercise Answers

1. $\frac{\sqrt{2}-\sqrt{6}}{4}$
2. $\frac{\sqrt{2}+\sqrt{6}}{4}$
3. $\frac{\sqrt{2}+\sqrt{6}}{4}$
4. $\frac{\sqrt{6}-\sqrt{2}}{4}$
5. $-2 + \sqrt{3}$
6. $-2 - \sqrt{3}$
7. $\frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x$
8. $-\frac{\sqrt{2}}{2} \sin x - \frac{\sqrt{2}}{2} \cos x$
9. $\frac{1}{2} \sin x - \frac{\sqrt{3}}{2} \cos x$
10. $-\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x$
11. $\sec t$
12. $\csc \theta$
13. $\tan x$
14. $\cot x$
15. $\sin(-3x)$
16. $\tan\left(\frac{x}{10}\right)$
17. $\sin(a+b) = \frac{-2-5\sqrt{3}}{12}; \cos(a+b) = \frac{\sqrt{5}+2\sqrt{15}}{12}$
18. $\sin(a+b) = \frac{4-6\sqrt{2}}{15}; \cos(a+b) = \frac{3-8\sqrt{2}}{15}$
19. $\frac{1}{2}$
20. $\frac{\sqrt{2}-\sqrt{6}}{4}$
21. $-\frac{\sqrt{3}}{3}$
22. $\frac{\sqrt{2}+\sqrt{6}}{4}$
23. $\frac{\sqrt{2}-\sqrt{6}}{4}$
24. $\frac{-\sqrt{2}-\sqrt{6}}{4}$
25. $\frac{\sqrt{2}+\sqrt{6}}{4}$
26. $-2 + \sqrt{3}$